

for programmers. Most contestants came from Bengaluru, Pune, Hyderabad, Chennai, New Delhi and Mumbai.

Specifics for development

"Engineers with multi-disciplinary skills are in high demand," explains Dr Milind Kulkarni, director - engineering institutes, ITM group. According to the market, Big Data analytics and Cloud skills are important because IoT sensors tend to send large amounts of data in real time. But the most important skills developers need are proficiencies in Cloud, mobile and data analytics. Expertise on the tri-fecta would govern salary and status with time.

Since the IoT requires dedicated hardware, there is a large requirement for developers with the knowledge of embedded systems. Since embedded hardware as well as software has to be continually updated, efficient development is now prioritised. Development skills will get you started, but creativity would keep you going.

Considering the hardware

Manufacturing hardware adds a certain level of complexity, so as a developer, refining some of those skills in hardware is likely to add to your skill set.

The IoT relates to connecting every device to a network. A very large number of these connections require sensors and dedicated hardware development. Current sensors include, "vibration, radio frequency, temperature, humidity, pressure and more," lists Hrishikesh Kamat, chief executive officer, Shalaka Technologies Pvt Ltd. A dedicated hardware learns from every recorded activity. "The system can learn and update after every activity," adds Kamat.

Automation also covers a lot

of opportunities in product- and service-oriented industries. According to Siddarth Bharwani, vice president, Jetking Infotrain Ltd, "Researchers, robotics system engineers, senior robotics specialists, analysts, trainers, robotics technicians, supervisors and assistant engineers in the design and application of robot systems comprise major opportunities."

A lot of confusion in software

Software development is an area that requires fine tuning. Once we have a higher number of devices, decision towards a particular technology can be made easily. Ac-

majority of the devices along with their Cloud and mobile components allowing passwords such as 1234 or 123456, security becomes a major concern.

Additionally, higher the number of connected devices, higher the opportunities to break the system. Last year saw a rise in the number of data attacks, which were more than a hundred thousand. Bharwani adds, "Enterprises have to protect themselves, their services and data from vulnerabilities." Hence, efficient security professionals would be in high demand. However, since most of the potential weak links are similar

to mobile or Web applications, people currently working on security have an edge; after all, anybody would bet on an experienced horse.

Difference in development now

At first glance, it does not feel like much of a difference. People are working on development and have been for more than 20 years. The biggest defining factor with the IoT is the things themselves.

Since there is no fixed device and range of development is unlimited, coding for this broad range of devices will define IoT software development. For hardware, most automated devices are connected and sensor-oriented. Hence, incorporating sensors in every imaginable device would be the change in development. These devices range from a simple wearable to dynamic streetlights with warnings for over-speeding, and would require developers to be creative in finding solutions for specific scenarios.

Available opportunities

Metros and big cities are becoming target spots for job roles.

JOB OPPORTUNITIES AT A GLANCE

By location	By technology	By programming language
Bengaluru	IoT hardware development	C
Pune	IoT software development	C++
New Delhi	Cyber security	Java
Hyderabad	Data scientist	Python
Chennai	Solutions architect	C#

cording to some industry experts, C programming, JavaScript and PHP skills should be in demand. These comprise the most basic in development and, hence, would provide a better way to work the system.

Some other experts believe that the IoT development could see demand in Java, Android, CSS and JavaScript skills. TechGig corroborates this with respect to language options. This sector is, however, limited to website development and management.

Cyber security experts

In a connected world, another matter of concern is security. According to a study by HP Fortify, conducted in 2015, "Six out of ten devices that provide user interfaces were vulnerable to a range of issues such as persistent XSS and weak credentials." If we are to envision a fully-connected world with